

INDUSLINK: An AI-Powered Platform for Bridging the Gap Between Undergraduates and the Industry

Project ID: 25-26J-281

Project Proposal Report

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INDUSLINK: AN AI-POWERED INDUSTRY -ACADEMIC SOCIAL FEED & ENGAGEMENT ANALYSIS

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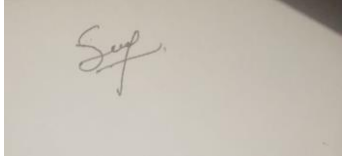
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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

The above candidate is carrying out research for the undergraduate dissertation under my supervision.

.....

Signature of the supervisor:

(Mr. Suresh Niroshan)

Date:

.....

Signature of the Co-supervisor:

(Ms. Fathima Fanoon)

Date:

Abstract

This work focuses on the design and analysis of an Industry-Academic Social Feed that bridges the interaction gap between student and professionals while generating actionable insights through data-driven techniques. Inspired by platforms such as LinkedIn, the system provides a dynamic feed where students, recruiters, and industry experts can share updates, job stories, career guidance, and inquiries. Engagement features such as likes, comments, shares, skill-tagging, and hashtags (e.g., #AI, #Internship) are incorporated to enhance discoverability and promote professional branding.

This research scope emphasizes AI-driven engagement analysis. Natural Language Processing (NLP) techniques, including sentiment analysis and topic modeling, are applied to extract underlying emotions, detect emerging themes, and cluster posts by relevance. Engagement analytics further measure user interactions through metrics such as interaction rate, post popularity, and reach, enabling a deeper understanding of behavioral patterns within the student–industry ecosystem. In addition, predictive models evaluate how posting activity influences student visibility and the likelihood of attracting recruitment opportunities.

By combining social media interaction patterns with advanced machine learning methods, this work contributes to developing a hybrid academic–professional networking environment. The outcomes aim to provide students with data-informed feedback on their personal branding, support institutions in identifying relevant career trends, and assist recruiters in recognizing emerging talent pools. Ultimately, the proposed system advances the integration of social networking, career development, and AI-powered analytics into a unified platform for fostering meaningful academic–industry engagement.

Keywords: Industry–academic engagement, social feed, sentiment analysis, topic modeling, natural language processing (NLP), engagement analytics, student visibility, recruitment insights, AI-driven networking, personal branding.

Table of Contents

Declaration of the Candidate & Supervisor	i
Abstract	ii
Table of Contents	iii
List of Figures	iv

List of Tables	v
List of Abbreviations	vi
1.0 Introduction	1
1.1 Background & Literature Survey	1
1.2 Research Gap	2
1.3 Research Problem	3
2.0 Objectives	4
2.1 Main Objective	4
2.2 Specific Objectives	4
3.0 Methodology	5
3.1 Data Collection & Preprocessing	5
3.2 NLP & Clustering Techniques	5
3.3 Proposed Approach & Task Breakdown	6
3.4 Work Breakdown Structure (WBS)	7
3.5 Gantt Chart & Project Plan	8
4. Project Requirements	9
4.1 Functional Requirements	9
4.2 Non-Functional Requirements	9
4.3 System/User Requirements	9
5. Budget and Budget Justification	10
6. Anticipated Results & Conclusion	11
7. Commercialization Potential	11
8. References	12
Appendices (if any)	12

List of Figures

Figure 3.3 Work Breakdown Structure (WBS)	7
Figure 3.4 Gantt Chart	8

List of Tables

Table 1.1 Research Gap Comparison2

Table 2.1 Budget Distribution10

List of Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
TF-IDF	Term Frequency-Inverse Document Frequency
BERT	Bidirectional Encoder Representations from Transformers
WBS	Work Breakdown Structure
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
SMEs	Small and Medium Enterprises

1. Introduction

My part of the IndusLink project focuses on developing the AI-powered feed, which forms the core of the platform’s interactive experience. This component leverages Natural Language Processing (NLP), sentiment analysis, and topic modeling techniques to analyze user-generated content, industry updates, and student interactions. The AI feed dynamically curates relevant posts, internship opportunities, and career guidance tailored to individual student profiles, ensuring that each user receives personalized and actionable insights.

By integrating engagement metrics and predictive analytics, the AI-powered feed enhances visibility, interaction, and knowledge sharing between students and industry professionals, effectively bridging the academic–industry gap and supporting informed career decisions.

1.1 Background & Literature Survey

The expansion of digital platforms and professional networking technologies has significantly transformed how students and industry professionals interact, share knowledge, and build career

opportunities. Traditional job portals and recruitment systems provide access to job postings, yet they lack interactive features that allow real-time engagement between undergraduates and industry stakeholders. Platforms such as LinkedIn have demonstrated the value of social feeds, where students and employers exchange updates, career tips, and recruitment opportunities. However, these global platforms are not optimized for academic–industry integration, nor do they provide localized insights into how engagement influences career development and recruitment.

Recent research highlights the growing importance of Artificial Intelligence (AI) in analyzing user-generated content within social networks. Studies show that Natural Language Processing (NLP) can be effectively applied for sentiment analysis to identify user opinions, emotions, and perceptions in social media discussions [1]. This enables organizations to better understand community responses and adjust their strategies accordingly. Additionally, topic modeling techniques such as Latent Dirichlet Allocation (LDA) and BERTopic have been widely applied to cluster social posts and detect emerging themes [2]. Such methods have proven useful in surfacing trending discussions in educational and professional networks, ensuring that users remain updated on relevant topics.

Engagement metrics—including likes, comments, shares, reach, and interaction rate—have been studied as key indicators of user influence and visibility. For example, research by Cvijikj and Michahelles [3] demonstrated that analyzing interaction patterns helps explain content popularity and can predict post performance on social platforms. Similarly, Liu et al. [4] emphasized the role of post engagement analysis in improving recommendation systems, showing that higher interaction correlates strongly with visibility and recruitment attention. Predictive models have further been developed to estimate post popularity and evaluate how user activity impacts professional opportunities [5].

In the academic–industry context, several works emphasize the role of online engagement in career development. Ghosh et al. [6] highlighted how online networking platforms help students build personal branding, while Ahmed and Siddiqui [7] found that employer engagement on digital platforms improves student recruitment outcomes. Despite these advances, there remains a gap in systems that integrate social feed dynamics with AI-driven engagement analytics, specifically designed for bridging academia and industry.

This research addresses that gap by developing a dynamic social feed platform tailored for students and professionals, integrating NLP-based sentiment analysis, topic modeling, and engagement analytics. By doing so, the system not only enhances interaction but also generates actionable insights to improve student visibility, guide institutional content strategies, and support recruiters in talent discovery.

1.2 Research Gap

While platforms such as LinkedIn, ResearchGate, and academic social networks provide spaces for professional engagement, they are primarily designed for global users and do not directly address the student–industry engagement gap in localized contexts such as Sri Lanka. Current platforms focus heavily on professional networking and job listings, but they often overlook the educational alignment between what students share (skills, projects, career aspirations) and what recruiters or HR professionals value in potential candidates.

Existing studies have explored social media analytics for understanding job market trends, such as sentiment analysis of career-related posts [1] and topic modeling to identify emerging industry demands [2]. However, these works mainly concentrate on macro-level insights rather than student-centered employability outcomes. In the Sri Lankan setting, studies [3][4][5] emphasize graduate unemployment and employer dissatisfaction but lack technical frameworks that analyze student visibility, post engagement, and skill-tagged social interaction in professional networks.

The research gap, therefore, lies in the absence of an AI-powered academic–industry social engagement platform that can:

- Collect and analyze student-HR generated social feed posts.
- Apply NLP techniques for sentiment and topic clustering of posts.
- Track engagement metrics (likes, shares, comments) to evaluate student visibility.
- Map skill-tagging and trending topics (#AI, #Internship, etc.) to recruitment likelihood.

Research Gap Feature	Smith et al. [1]	Narula et al. [2]	Local SriLankan Employment Studies [3][4][5]	Proposed Research Solution
Sentiment analysis of career-related posts	✓	✗	✗	✓
NLP-based topic modeling for career/industry trends	✓	✓	✗	✓
Engagement analysis (likes,shares,comments)	✗	✗	✗	✓
Influence of posting activity on recruitment/visibility	✗	✗	✓ (employment issues only)	✓
Focus on Sri Lankan student–industry gap	✗	✗	✗	✓

Table 1.1 Research Gap Comparison

1.3 Research Problem

In Sri Lanka, IT undergraduates face challenges in connecting effectively with industry professionals, which limits their employability and career growth. Existing social and professional platforms allow students to post updates, like, comment, or share content, but they do not provide any analysis of engagement or insights on how students' posts affect their visibility and career prospects. This leads to information overload and missed opportunities, as students struggle to identify content or discussions that are relevant to their skills and interests.

Additionally, there is no system to analyze trends, sentiment, or engagement in student-industry interactions, leaving valuable data underutilized. Therefore, there is a critical need to develop an AI-driven social feed that not only connects students with industry professionals but also applies Natural Language Processing (NLP) and engagement analytics to provide actionable insights. Such a system would help students optimize their online presence, understand emerging industry trends, and improve alignment between their academic skills and market demands.

2. Objectives

2.1 Main Objective

To design and implement a dynamic, AI-driven Industry–Academic Social Feed that not only facilitates structured interactions between IT undergraduates and industry professionals but also leverages Natural Language Processing (NLP), topic modeling, sentiment analysis, and engagement metrics to provide actionable insights, identify emerging industry trends, and enhance student visibility, personal branding, and employability within the Sri Lankan IT job market

2.2 Specific Objectives

The research aims to develop a comprehensive Industry–Academic Social Feed that connects IT undergraduates with industry professionals while providing actionable insights through AI-driven analytics. To achieve this, the study will first design and implement a social feed interface similar to LinkedIn, allowing users to post updates, like, comment, share, and tag skills or industry-relevant topics. The platform will leverage Natural Language Processing (NLP) techniques to perform sentiment analysis on posts, helping to understand the attitudes and perceptions of students and industry participants toward career and professional development. Additionally, topic modeling algorithms such as LDA and BERTopic will be applied to cluster posts by themes, emerging skills, and industry trends, enabling the identification of relevant discussions and popular topics. Engagement metrics, including likes, shares, comments, post reach, and interaction rates, will be analyzed to evaluate visibility and influence within the network. The research will also develop a recommendation system that provides students with actionable feedback on improving their visibility, personal branding, and alignment with industry expectations. Finally, the study will assess the impact of posting activity on career-related outcomes, including networking opportunities, skill recognition, and recruitment likelihood, with all analyses adapted to the Sri Lankan IT context to ensure local relevance for students, universities, and industry stakeholders.

3. Methodology

This research adopts a mixed-methods approach, combining data collection, Natural Language Processing (NLP), topic modeling, engagement analytics, and system development to bridge the gap between IT undergraduates and industry professionals in Sri Lanka. The methodology consists of several key stages.

3.1 Data Collection & Preprocessing

The first step of this research involves collecting relevant data from multiple sources to capture student–industry interactions in the Sri Lankan IT sector. Primary sources include social media platforms such as LinkedIn, Facebook, and Twitter, where students and industry professionals share updates, job opportunities, and career advice. Secondary sources include online job portals (TopJobs, company websites) to obtain IT job postings and skill requirements. The collected data will be preprocessed using standard NLP techniques to ensure consistency and usability. Preprocessing steps include text cleaning, tokenization, lemmatization, stop-word removal, and translation of posts in local languages (Sinhala and Tamil) where applicable. The goal is to convert unstructured text data into a structured format suitable for analysis and modeling.

3.2 NLP & Clustering Techniques

Natural Language Processing (NLP) forms the core of this research for analyzing content and extracting meaningful patterns. Sentiment analysis will classify posts into positive, negative, or neutral categories, providing insights into student and industry attitudes. Topic modeling algorithms, such as Latent Dirichlet Allocation (LDA) and BERTopic, will cluster posts into coherent themes, such as trending skills, industries, or internship opportunities. In addition, semantic clustering of job descriptions will be performed using sentence embeddings to group similar roles, even if the keyword descriptions differ. This approach enables a deeper understanding of content trends, student concerns, and emerging skill demands.

3.3 Proposed Approach & Task Breakdown

The proposed approach combines a social feed interface with AI-driven analytics. Key tasks include:

Interface Development: Create a LinkedIn-style feed allowing posts, likes, comments, shares, and tagging of skills or industries.

Data Aggregation & Preprocessing: Collect data from job portals and social media, and apply NLP preprocessing.

NLP Analysis: Perform sentiment analysis and topic modeling to detect themes and emerging trends.

Engagement Analytics: Analyze likes, shares, comments, and reach to measure visibility and influence.

Recommendations: Provide actionable feedback for students to optimize their online presence and career visibility.

Validation: Collect feedback from students and industry professionals to assess system effectiveness.

3.4 Work Breakdown Structure (WBS)

The WBS for this research is structured in hierarchical phases:

Phase 1 – Data Collection: Social media scraping, job portal extraction, preprocessing.

Phase 2 – NLP & Analytics: Sentiment analysis, topic modeling, semantic clustering.

Phase 3 – System Development: Design social feed interface, integrate NLP modules, implement recommendation engine.

Phase 4 – Evaluation: Measure engagement metrics, validate sentiment and topic analysis, collect user feedback.

Phase 5 – Reporting: Prepare dashboards, visualize trends, document results, and provide insights for students and industry stakeholders.

This WBS ensures systematic progress, with clear deliverables and timelines for each phase.

Research Project WBS - Gantt Chart

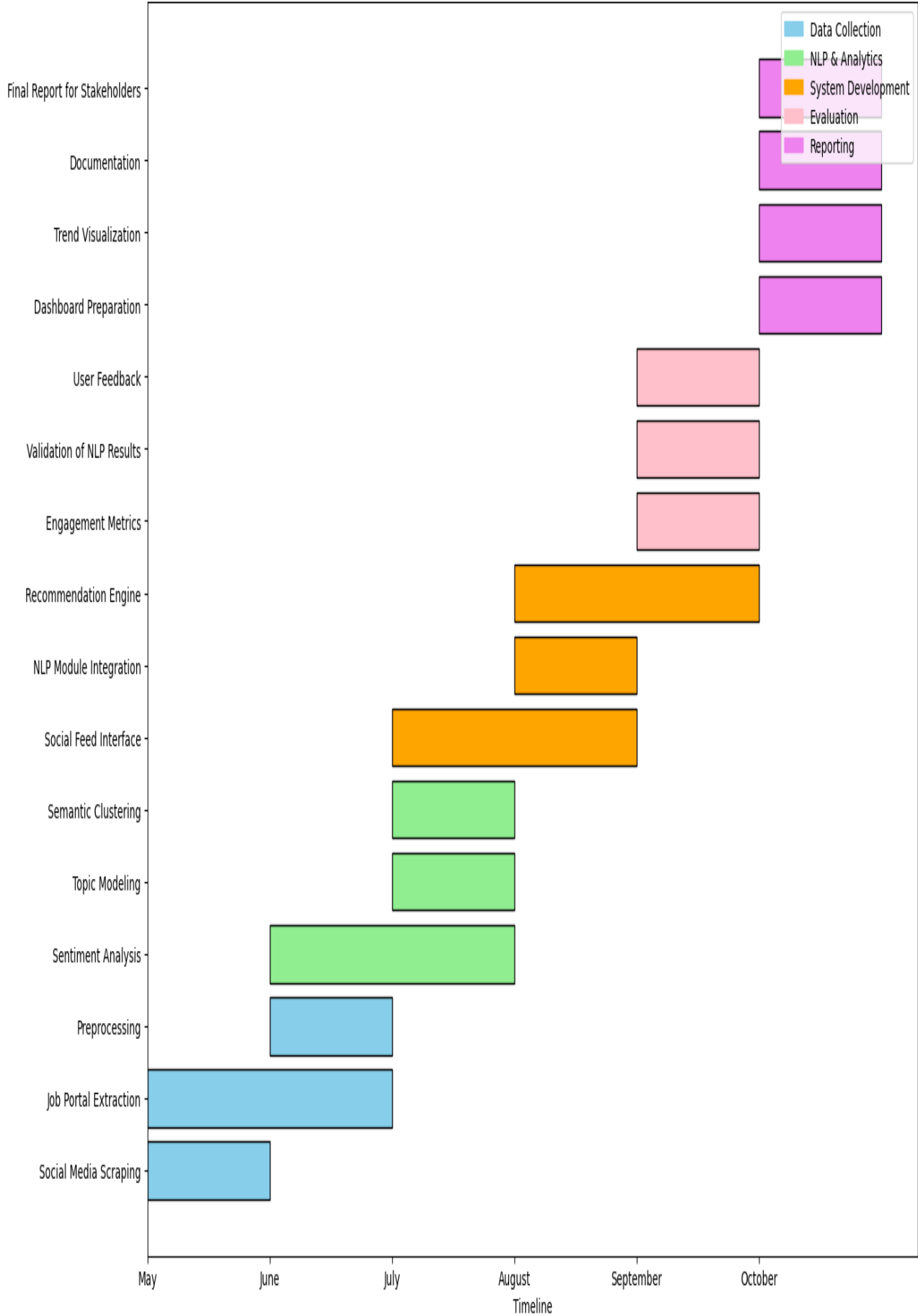


Figure 3.4 Work Breakdown Structure (WBS)

3.5 Grantt Chart & Project Plan

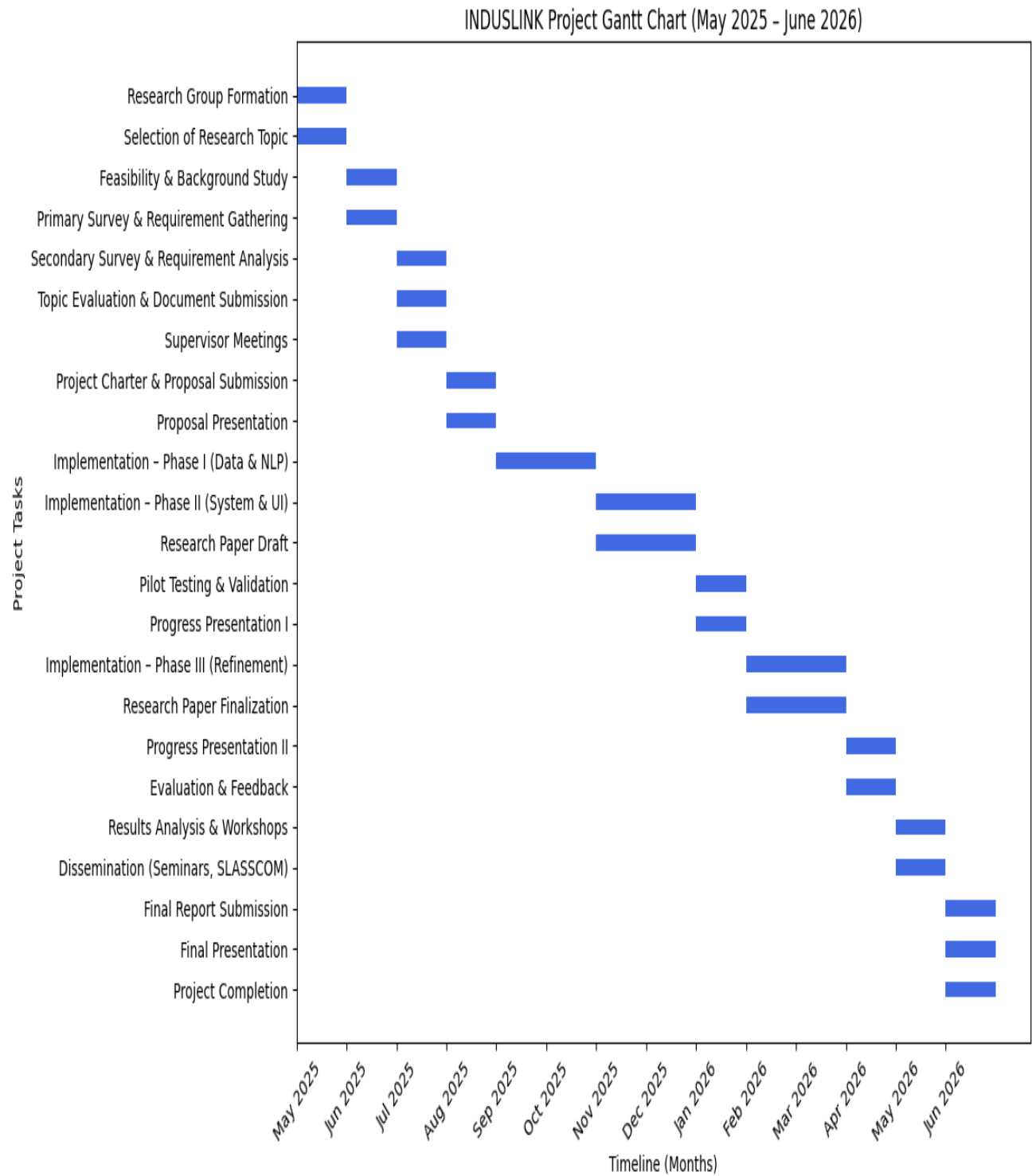


Figure 3.5 Gantt Chart

4. Project Requirements

4.1 Functional Requirements

- **User Registration and Profiles** — Students, recruiters, and administrators must be able to create, edit, and manage profiles securely.
- **Social Feed Interaction** — Users can create posts, comment, like, share, and tag skills or hashtags within a LinkedIn-style feed.
- **Job and Student Clustering** — The system must cluster job postings into categories and group students into personas based on skills.
- **Skill Gap Analysis** — The system should calculate and display personalized skill-gap reports for each student.
- **Recommendation Engine** — Provide students with suggestions on skills to acquire, hashtags to use, and optimal post timing for visibility.
- **Analytics Dashboards** — Students see visibility scores and branding insights; recruiters view skill heatmaps and candidate discovery tools.
- **Pilot Management Module** — Enable administrators to manage pilot participants, data collection, and evaluation.

4.2 Non-Functional Requirements

- **Performance** — System responses (feed refresh, recommendation queries) should complete within 2 seconds under normal load.
- **Scalability** — The platform must support at least 1,000 concurrent users during the pilot, with potential scale to 10,000+.
- **Security** — Ensure data encryption in transit (HTTPS/TLS) and at rest. Implement role-based access control and anonymization.
- **Usability** — Interfaces must be intuitive, mobile-responsive, and accessible, following WCAG 2.1 guidelines.
- **Reliability** — Maintain >95% uptime during the pilot phase with automated backup and recovery procedures.
- **Maintainability** — Codebase should follow modular architecture with documentation, unit tests, and continuous integration.
- **Explainability & Ethics** — AI recommendations must include transparent explanations and allow user feedback to refine outputs.

4.3 System/User Requirements

- **Students:** Create posts, access recommendations, view dashboards, and participate in mock interviews.
- **Recruiters:** Browse talent pools, view student skill profiles, and receive insights from dashboards.

- **Administrators:** Manage user accounts, oversee pilot testing, generate evaluation reports, and ensure compliance with ethical standards.

5. Budget and Budget Justification

Data Collection & preprocessing	• Instead of large paid APIs, freely available sources (government portals, LinkedIn job posts, university career pages) will be used. • Budget is for occasional API subscriptions and student survey data collection (printing, incentives, etc.)	LKR30000
Computing Resources	• High-cost GPU servers will be avoided. The project will rely on Google Colab (free GPU), university lab machines, and limited cloud credits. • This keeps training costs minimal while still allowing model experimentation	LKR40000
System Development	• Covers only essential features: a prototype social feed, clustering module, and analytics dashboard. • Development will use open-source frameworks (Python, TensorFlow/PyTorch, React) and free database tools (MongoDB Atlas free tier)	LKR 50000

Testing & Validation	• Focus groups will be limited to one faculty and a small recruiter network. • Budget covers survey printing, refreshments, and incentives for pilot participants	LKR 10000
Documentation	Covers report printing, binding, and sharing results	LKR 10000
Total estimate cost	LKR 140000	

Table 2.1 Budget Distribution

6. Anticipated Conclusion

This research anticipates that the development and pilot implementation of INDUSLINK will demonstrate the viability of an AI-powered platform in addressing Sri Lanka’s persistent gap between academic training and industry expectations. The system is expected to show that transformer-based sentiment analysis and topic modeling can effectively extract insights from student and employer data, while clustering methods can meaningfully group skills and job categories. By integrating these modules into a social feed and analytics dashboard, INDUSLINK will provide undergraduates with actionable recommendations to enhance visibility, improve personal branding, and identify skill gaps.

It is anticipated that students participating in the pilot will demonstrate measurable improvements in their employability readiness, as reflected in higher engagement with recruiters, increased awareness of required skills, and improved confidence in professional self-presentation. Recruiters, in turn, are expected to benefit from clearer insights into emerging talent pools and skill trends, thus validating the system’s dual relevance.

The project aims to conclude that a localized, AI-driven employability platform can complement traditional career guidance mechanisms and serve as a scalable model for other developing countries facing similar challenges. Ultimately, the findings are expected to contribute not only to academic literature on employability and AI in education but also to practical industry solutions, thereby ensuring both scholarly and societal impact

7. Commercialization Potential

INDUSLINK is commercially viable because it targets a real, unmet need in student–industry engagement, leverages AI for actionable insights, and offers multiple revenue streams (B2C

subscriptions, B2B analytics, sponsorships). Its scalability, localization, and AI-driven uniqueness make it attractive for investors, universities, and recruiters alike.

8. References

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Appendices

[If any, include here.]